

Project Report

Project Title: Smart Lighting System Using LDR Sensor

Objective

The objective of this project is to design a smart lighting system that automatically controls an LED based on the surrounding light intensity using an LDR (Light Dependent Resistor) and Arduino UNO.

Components Used

- Arduino UNO
- LDR (Light Dependent Resistor)
- LED
- Resistor
- Breadboard
- Jumper Wires

Working Principle

The LDR senses the intensity of light in the environment and sends an analog signal to the Arduino. The Arduino continuously reads the sensor value through the analog input pin. When the light intensity is above the threshold value, the system identifies it as light. When the intensity falls below the threshold value, the system identifies it as darkness. The status is displayed on the Serial Monitor, and the LED can be controlled automatically based on the light conditions.

Code

```
#define led 13
```

```
#define ldr A0
```

```
void setup() {
```

```
  Serial.begin(9600);
```

```
  pinMode(led, OUTPUT);
```

```
  pinMode(ldr, INPUT);
```

```
}
```

```
void loop() {
```

```
  int ldr_value = analogRead(ldr); // Read sensor value (0–1023)
```

```
  Serial.println(ldr_value);
```

```
  // Dark room => low value, turn LED on
```

```
  if (ldr_value < 500) {
```

```
    digitalWrite(led, HIGH); // LED ON

} else {

    digitalWrite(led, LOW); // LED OFF

}

}
```

Code Explanation

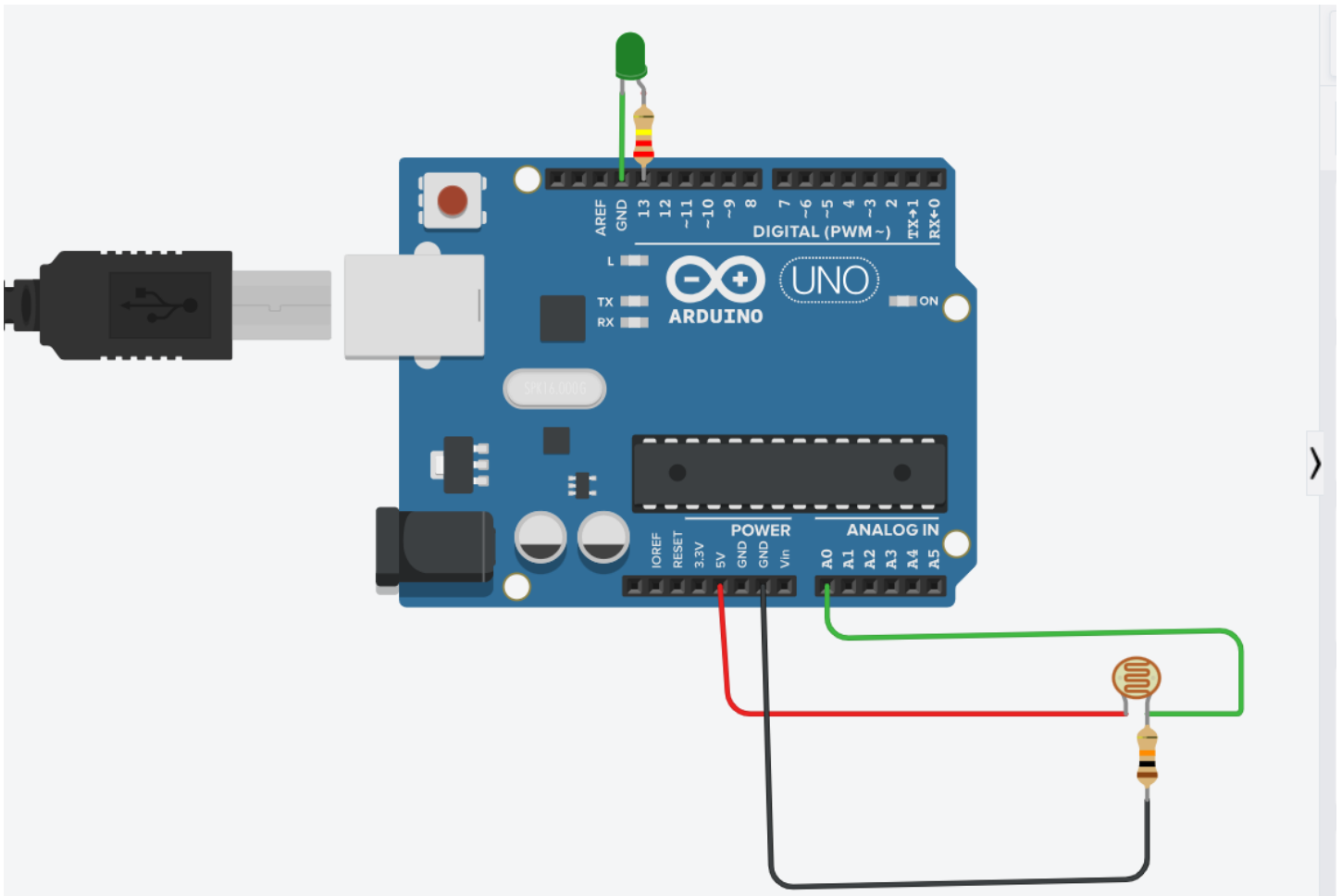
The program is designed to create a Smart Lighting System using an LDR (Light Dependent Resistor) and an Arduino UNO. The LDR is connected to analog pin **A0**, and the LED is connected to digital pin **13**.

In the `setup()` function, serial communication is initialized at **9600 baud rate** to display sensor values on the Serial Monitor. The LED pin is configured as an output, and the LDR pin is configured as an input.

In the `loop()` function, the Arduino continuously reads the light intensity from the LDR using the `analogRead()` function. The sensor value ranges from **0 to 1023** and is stored in the variable `ldr_value`. This value is displayed on the Serial Monitor using `Serial.println()`.

The program then checks whether the sensor value is less than **500**. If the value is below 500, it indicates a dark environment, and the LED is turned ON using `digitalWrite(led, HIGH)`. If the value is 500 or higher, it indicates sufficient light, and the LED is turned OFF using `digitalWrite(led, LOW)`.

Thus, the system automatically controls the LED based on the surrounding light intensity, making it suitable for automatic lighting applications such as street lights and smart home systems.



Applications

- Automatic street lighting systems
- Smart home automation
- Garden and pathway lighting
- Energy-saving lighting systems
- Security and outdoor lighting

Conclusion

This project demonstrates the use of an LDR sensor with Arduino UNO to detect light intensity. The system can identify light and dark conditions and can be used for automatic lighting applications. It is a simple and effective example of an embedded system used in everyday life.